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(54) CRYSTAL FORM OF BETA-LACTAMASE INHIBITOR AND PREPARATION METHOD THEREFOR

KRISTALLINE FORM VON BETA-LACTAMASE-INHIBITOR UND HERSTELLUNGSVERFAHREN DAFÜR

FORME CRISTALLINE D'UN INHIBITEUR DE BETA-LACTAMASE ET SON PROCÉDÉ DE PRÉPARATION

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(56) References cited:
WO-A1-2014/091268
WO-A1-2017/206947
CN-A- 104 768 951
US-A1- 2014 288 051

Remarks:
The file contains technical information submitted after the application was filed and not included in this specification

EP 3 719 020 B1

Description

[0001] This application claims the benefit of the following application:

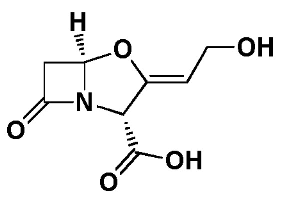
[0002] CN201711251386.3, filing date: December 01, 2017.

Field of the invention

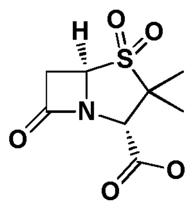
[0003] The present disclosure relates to a novel class of β-lactamase inhibitors, specifically discloses a compound of formula (I) or a pharmaceutically acceptable salt thereof.

Background of the invention

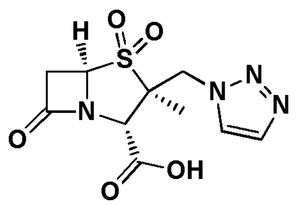
[0004] There are several mechanisms by which bacteria develop resistance to β-lactam antibiotics, and one of the principal mechanisms is the production of enzymes that can hydrolyze the β-lactam ring and thus inactivate the antibiotics. Bacteria can also selectively alter the target of antibiotics. For example, methicillin-resistant *Staphylococcus aureus* has developed multiple resistance which is associated with the production of new PBP_{2a}, increased synthesis of PBPs, and decreased drug affinity. β-lactamase can rapidly bind to certain enzyme-resistant β-lactam antibiotics, allowing the drug to remain in the periplasmic space of the cytoplasmic membrane and fail to reach the target to exert an antibacterial effect. In addition, the outer membrane of G-bacteria is not easily permeable to certain β-lactam antibiotics, resulting in non-specific low-level resistance. There are also some active exocytosis systems on the cytoplasmic membrane of bacteria, by which bacteria actively release drugs to the exterior. Therefore, the combination of a β-lactam antibiotic and a β-lactamase inhibitor is the most clinically effective method. Bacteria can produce various types of β-lactamases, which can be classified into four classes: A, B, C, and D according to their amino acid and nucleotide sequences. Classes A, B, and D enzymes catalyze hydrolysis with serine as an active site, and class B enzymes cleave the ring by one or more metal atoms at the active site.



Clavulanic acid



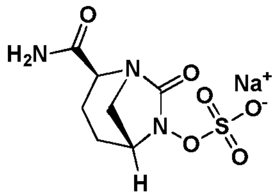
Sulbactam



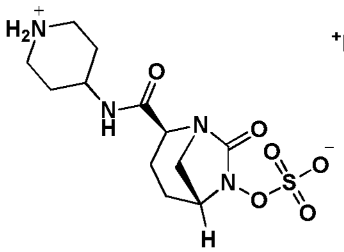
Tazobactam

[0005] The first well-known high-activity β-lactamase inhibitor is potassium clavulanate, and its combination with amoxicillin is still hot in the market to date. Two other important β-lactamase inhibitors on the market are sulbactam and tazobactam. These three drugs have a highly active β-lactam ring in their structures in common, which is the active site of these inhibitors. Although these three drugs are hot in the market, their antibacterial spectrum is very narrow. They are only effective on classes A and D β-lactamases, but are completely ineffective on class C enzymes and KPC enzymes which play an important role in class A enzymes.

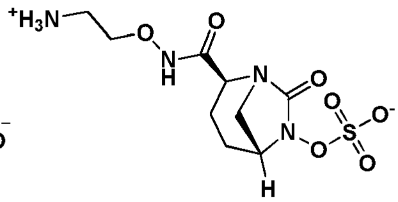
[0006] In February 2015, FDA approved a new β-lactamase inhibitor named avibactam (NXL-104). This drug containing a novel diazabicyclo structure has a broader antibacterial spectrum than those three previous generation β-lactamase inhibitors described above. The patents for β-lactamase inhibitors including WO2009133442, WO2009091856, WO2010126820, WO2012086241, WO2013030733, WO2013030735, WO2013149121, WO2013149136, WO2013180197, WO20140191268, WO2014141132, WO2014135931, WO2015063653, WO2015110885 and US20140296526, have disclosed a large number of new diazabicyclo compounds, among which MK-7655 and OP-0595 are two new drugs that have entered the clinical trial stage. MK-7655 has entered Phase III clinical trial stage, and OP-0595 has entered Phase I clinical trial stage. OP-0595 has excellent *in vitro* activity and is owned by Roche Pharmaceuticals. Therefore, diazabicyclo inhibitors will be a new direction for the development of β-lactamase inhibitors.



(NXL-104)
Avibactam



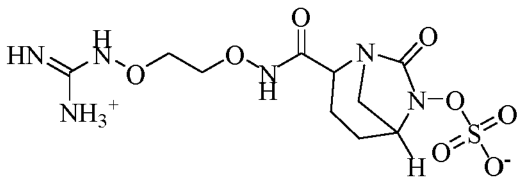
(MK-7655)



(OP-0595)

Content of the invention

[0007] A crystal form A of the compound of formula (I), wherein the X-ray powder diffraction pattern thereof comprises characteristic diffraction peaks at the following angle 2θ: 16.053 ± 0.2°, 16.53 ± 0.2°, 22.782 ± 0.2° and 25.742 ± 0.2°.



(I)

[0008] A crystal form A of the compound of formula (I), wherein the X-ray powder diffraction pattern thereof comprises characteristic diffraction peaks at the following angle 2θ: 16.053 ± 0.2°, 16.53 ± 0.2°, 18.501 ± 0.2°, 21.302 ± 0.2°, 21.778 ± 0.2°, 22.782 ± 0.2°, 25.742 ± 0.2° and 27.833 ± 0.2°.

[0009] In some embodiments of the present disclosure, the XRPD pattern of the crystal form A is as shown in Fig. 1.

[0010] In some embodiments of the present disclosure, the analytical data of the XRPD pattern of the crystal form A is as shown in Table 1.

Table 1

No.	2θ Angle (°)	d-spacing (Å)	Intensity	No.	2θ Angle (°)	d-spacing (Å)	Intensity
1	8.989	9.8297	387	18	28.997	3.0768	373
2	14.363	6.1618	143	19	29.366	3.0389	409
3	16.053	5.5164	1532	20	29.829	2.9928	108
4	16.53	5.3584	1785	21	30.632	2.9161	285
5	17.828	4.9712	151	22	31.388	2.8477	112
6	18.121	4.8914	489	23	32.351	2.7651	439
7	18.501	4.7918	625	24	32.989	2.713	112
8	20.177	4.3973	1185	25	33.334	2.6857	166
9	21.302	4.1676	1199	26	33.693	2.6579	130
10	21.778	4.0776	1255	27	34.575	2.5921	54
11	22.782	3.9001	1582	28	35.072	2.5565	130
12	23.889	3.7219	101	29	35.408	2.533	177
13	24.536	3.6251	462	30	36.287	2.4736	50
14	25.212	3.5294	384	31	37.045	2.4247	66
15	25.742	3.458	2124	32	37.539	2.3939	91
16	27.833	3.2028	1026	33	38.562	2.3328	91
17	28.323	3.1485	134	34	38.981	2.3086	47

Fig. 4 is the DVS pattern of the crystal form A of the compound of formula (I).

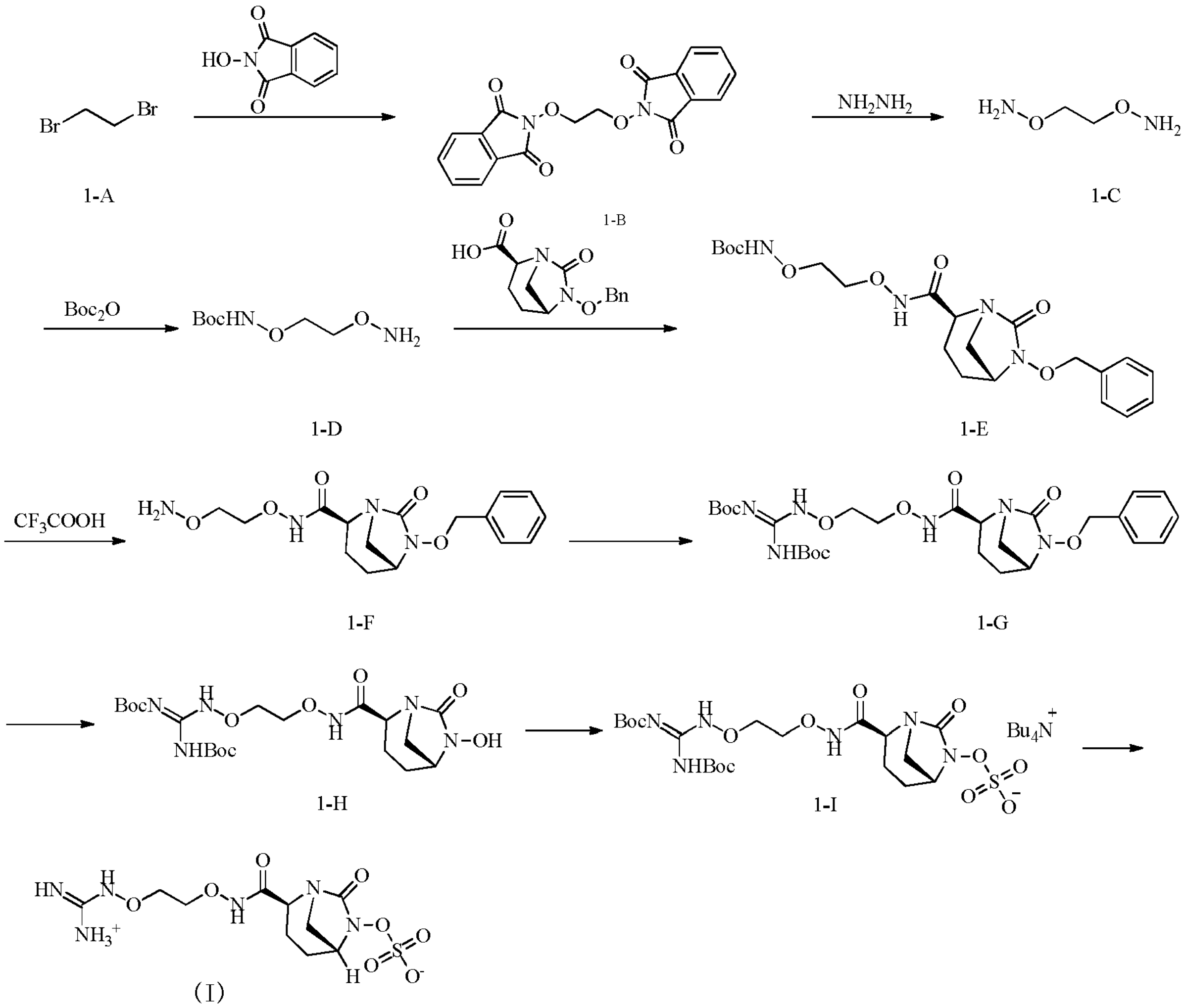
Fig. 5 shows the test results of the compound of formula (I) on KPC type β-lactamase producing *Klebsiella Pneumoniae.*

Detailed description of the preferred embodiment

[0033] In order to better understand the content of the present disclosure, the following embodiments further illustrate the present disclosure, but the present disclosure is not limited thereto.

Embodiment 1: Synthesis of the compound of formula (I)

[0034]

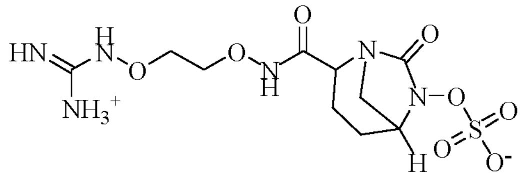


Step 1:

[0035] Starting material 1-A (50 g, 26.62 mmol), *N*-hydroxyphthalimide (8.69 g, 53.24 mmol) and triethylamine (6.73 g, 66.55 mmol) were dissolved in 100 mL *N,N*-dimethylformamide. The reaction solution was heated to 50 °C and stirred for 16 hours. The reaction solution was cooled to room temperature, poured into 100 mL ice-water under stirring, and filtered by suction. The obtained solid was washed with 10 mL cold water for three times, and dried to obtain compound 1-B.

Step 2:

[0036] The compound 1-B (6.0 g, 17.03 mmol) was suspended in 400 mL of dichloromethane and 150 mL of methanol,



(I)

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2. The crystal form A of the compound of formula (I) as defined in claim 1, wherein the X-ray powder diffraction pattern thereof comprises characteristic diffraction peaks at the following angle 2θ: 16.053 ± 0.2°, 16.53 ± 0.2°, 18.501 ± 0.2°, 21.302 ± 0.2° , 21.778 ± 0.2°, 22.782 ± 0.2°, 25.742 ± 0.2° and 27.833 ± 0.2°.

3. The crystal form A of the compound of formula (I) as defined in claim 2, wherein the XRPD pattern thereof is as shown in Fig. 1.

4. The crystal form A of the compound of formula (I) as defined in any one of claims 1 to 3, wherein the differential scanning calorimetry curve thereof has an exothermic peak at 221.11 ± 3 °C.

5. The crystal form A of the compound of formula (I) as defined in claim 4, wherein the DSC pattern thereof is as shown in Fig. 2.

6. The crystal form A of the compound of formula (I) as defined in any one of claims 1 to 3, wherein the thermogravimetric analysis curve thereof has a weight loss of 0.5689% occurred at 194.61 ± 3 °C.

7. The crystal form A of the compound of formula (I) as defined in claim 6, wherein the TGA pattern thereof is as shown in Fig. 3.

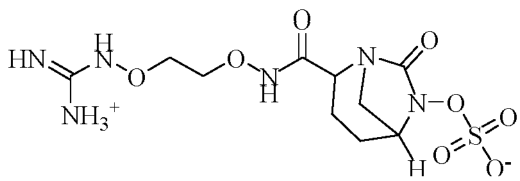
8. The crystal form A of the compound of formula (I) as defined in any one of claims 1 to 3, wherein the DVS pattern thereof is as shown in Fig. 4.

9. The crystal form A of the compound of formula (I) as defined in any one of claims 1 to 3, wherein the crystal form A of the compound of formula (I) has a hygroscopic weight gain of 0.2910% at 25 °C/80% RH

10. A method for preparing a crystal form A of the compound of formula (I) as defined in claim 1, comprising:

(a) adding the compound of formula (I) to a solvent and heating to 55-60 °C until it is completely dissolved;
(b) slowly cooling to 0 °C under stirring;
(c) stirring for 10-16 hours for crystallization;
(d) filtering, and drying by suction;

wherein the solvent is pure water;



(I)

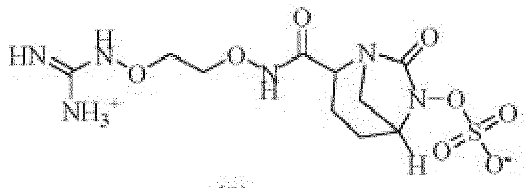
11. The crystal form A of the compound of formula (I) as defined in any one of claims 1-9 or the crystal form A of the compound of formula (I) prepared by the method as defined in claim 10 for use as a β-lactamase inhibitor for treating bacterial infection.

12. The crystal form A of the compound of formula (I) as defined in any one of claims 1-9 or the crystal form A of the compound of formula (I) prepared by the method as defined in claim 10 for use as a β-lactamase inhibitor for treating infection of carbapenemase-producing bacteria.

13. The crystal form A of the compound of formula (I) as defined in any one of claims 1-9 or the crystal form A of the compound of formula (I) prepared by the method as defined in claim 10 for use as a β-lactamase inhibitor for treating infection of KPC-2 carbapenemase-producing *Klebsiella Pneumoniae,* NDM-1 carbapenemase-producing *Klebsiella Pneumoniae,* or OXA-181 carbapenemase-producing *Klebsiella Pneumoniae.*

Patentansprüche

1. Kristalline Form A der Verbindung der Formel (I), wobei das Röntgen-Pulverdiffraktogramm (XRPD) davon charakteristische Beugungsmaxima bei dem folgenden Winkel 2θ umfasst: 16,053 ± 0,2°, 16,53 ± 0,2°, 22,782 ± 0,2° und 25,742 ± 0,2°,



(I)

2. Kristalline Form A der Verbindung der Formel (I), wie in Anspruch 1 definiert wird, wobei das Röntgen-Pulverdiffraktogramm davon charakteristische Beugungsmaxima bei dem folgenden Winkel 2θ umfasst: 16,053 ± 0,2°, 16,53 ± 0,2°, 18,501 ± 0,2°, 21,302 ± 0,2°, 21,778 ± 0,2°, 22,782 ± 0,2°, 25,742 ± 0,2° und 27,833 ± 0,2°.

3. Kristalline Form A der Verbindung der Formel (I), wie in Anspruch 2 definiert wird, wobei das XRPD-Muster davon so ist, wie in Fig. 1 gezeigt wird.

4. Kristalline Form A der Verbindung der Formel (I), wie in irgendeinem der Ansprüche 1 bis 3 definiert wird, wobei die dynamische Differenzkalorimetrie (DSC) davon ein exothermes Maximum bei 221,11 ± 3°C aufweist.

5. Kristalline Form A der Verbindung der Formel (I), wie in Anspruch 4 definiert wird, wobei das DSC-Muster davon so ist, wie in Fig. 2 gezeigt wird.

6. Kristalline Form A der Verbindung der Formel (I), wie in irgendeinem der Ansprüche 1 bis 3 definiert wird, wobei die thermogravimetrische Analysekurve davon einen Gewichtsverlust von 0,5689% aufweist, aufgetreten bei 194,61 ± 3°C.

7. Kristalline Form A der Verbindung der Formel (I), wie in Anspruch 6 definiert wird, wobei das TGA-Muster davon so ist, wie in Fig. 3 gezeigt wird.

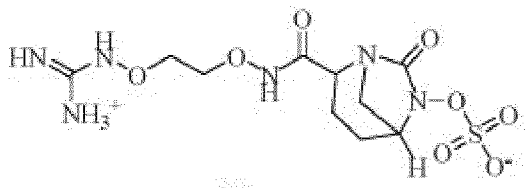
8. Kristalline Form A der Verbindung der Formel (I), wie in irgendeinem der Ansprüche 1 bis 3 definiert wird, wobei das DVS-Muster davon so ist, wie in Fig. 4 gezeigt wird.

9. Kristalline Form A der Verbindung der Formel (I), wie in irgendeinem der Ansprüche 1 bis 3 definiert wird, wobei die kristalline Form A der Verbindung der Formel (I) einen hygroskopischen Gewichtszuwachs von 0,2910% bei 25°C/80% RH aufweist.

10. Verfahren zur Herstellung einer kristallinen Form A der Verbindung der Formel (I), wie in Anspruch 1 definiert wird, umfassend:

(a) Zugabe der Verbindung der Formel (I) zu einem Lösungsmittel und Erhitzung auf 55-60°C bis sie vollständig aufgelöst ist;
(b) langsame Abkühlung auf 0°C unter Rühren;
(c) Rühren für 10-16 Stunden für Kristallisierung;
(d) Filtrierung und Saugtrockung;

wobei das Lösungsmittel reines Wasser ist;



(I)

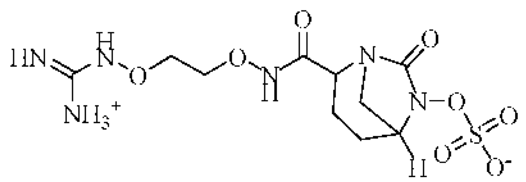
11. Kristalline Form A der Verbindung der Formel (I), wie in irgendeinem der Ansprüche 1 bis 9 definiert wird, oder die kristalline Form A der Verbindung der Formel (I), hergestellt durch das Verfahren, wie es in Anspruch 10 definiert wird, zur Verwendung als ein β-Lactamase-Inhibitor zur Behandlung einer bakteriellen Infektion.

12. Kristalline Form A der Verbindung der Formel (I), wie in irgendeinem der Ansprüche 1 bis 9 definiert wird, oder die kristalline Form A der Verbindung der Formel (I), hergestellt durch das Verfahren, wie es in Anspruch 10 definiert wird, zur Verwendung als ein β-Lactamase-Inhibitor zur Behandlung einer Infektion mit Carbapenemase-produzierenden Bakterien.

13. Kristalline Form A der Verbindung der Formel (I), wie in irgendeinem der Ansprüche 1 bis 9 definiert wird, oder die kristalline Form A der Verbindung der Formel (I), hergestellt durch das Verfahren, wie es in Anspruch 10 definiert wird, zur Verwendung als ein β-Lactamase-Inhibitor zur Behandlung einer Infektion mit KPC-2-Carbapenemase-produzierenden *Klebsiella Pneumoniae,* NDM-1-Carbapenemase-produzierenden *Klebsiella Pneumoniae* oder OXA-181-Carbapenemase-produzierenden *Klebsiella Pneumoniae.*

Revendications

1. Forme cristalline A du composé de formule (I), dans laquelle le diagramme de diffraction des rayons X sur poudre de celle-ci comprend des pics de diffraction caractéristiques à l'angle 2θ suivant : 16,053 ± 0,2°, 16,53 ± 0,2°, 22,782 ± 0,2° et 25,742 ± 0,2°,



(I)

2. Forme cristalline A du composé de formule (I) telle que définie dans la revendication 1, dans laquelle le diagramme de diffraction des rayons X sur poudre de celle-ci comprend des pics de diffraction caractéristiques à l'angle 2θ suivant : 16,053 ± 0,2°, 16,53 ± 0,2°, 18,501 ± 0,2°, 21,302 ± 0,2°, 21,778 ± 0,2°, 22,782 ± 0,2°, 25,742 ± 0,2° et 27,833 ± 0,2°.

3. Forme cristalline A du composé de formule (I) telle que définie dans la revendication 2, dans laquelle le diagramme XRPD de celle-ci est tel que représenté sur la figure 1.

4. Forme cristalline A du composé de formule (I) telle que définie dans l'une quelconque des revendications 1 à 3, dans laquelle la courbe de calorimétrie à balayage différentiel de celle-ci présente un pic exothermique à 221,11 ± 3 °C.

5. Forme cristalline A du composé de formule (I) telle que définie dans la revendication 4, dans laquelle le diagramme DSC de celle-ci est tel que représenté sur la figure 2.

6. Forme cristalline A du composé de formule (I) telle que définie dans l'une quelconque des revendications 1 à 3, dans laquelle la courbe d'analyse thermogravimétrique de celle-ci présente une perte de poids de 0,5689 % se produisant à 194,61 ± 3 °C.

7. Forme cristalline A du composé de formule (I) telle que définie dans la revendication 6, dans laquelle le diagramme TGA de celle-ci est tel que représenté sur la figure 3.

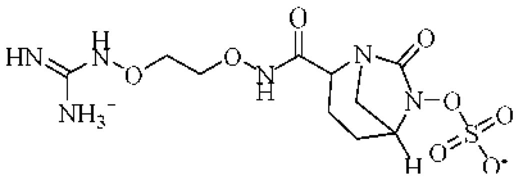
8. Forme cristalline A du composé de formule (I) telle que définie dans l'une quelconque des revendications 1 à 3, dans laquelle le diagramme DVS de celle-ci est tel que représenté sur la figure 4.

9. Forme cristalline A du composé de formule (I) telle que définie dans l'une quelconque des revendications 1 à 3, dans laquelle la forme cristalline A du composé de formule (I) présente un gain de poids hygroscopique de 0,2910 % à 25 °C/HR de 80 %.

10. Procédé de préparation d'une forme cristalline A du composé de formule (I) telle que définie dans la revendication 1, comprenant les étapes consistant à :

(a) ajouter le composé de formule (I) à un solvant et chauffer à 55-60 °C jusqu'à ce qu'il soit complètement dissous ;
(b) refroidir lentement jusqu'à 0 °C sous agitation ;
(c) agiter pendant 10-16 heures à des fins de cristallisation ;
(d) filtrer, et sécher par aspiration ;

dans lequel le solvant est de l'eau pure ;



(I)

11. Forme cristalline A du composé de formule (I) telle que définie dans l'une quelconque des revendications 1-9 ou forme cristalline A du composé de formule (I) préparée par le procédé tel que défini dans la revendication 10 pour une utilisation comme inhibiteur de β-lactamase pour traiter une infection bactérienne.

12. Forme cristalline A du composé de formule (I) telle que définie dans l'une quelconque des revendications 1-9 ou forme cristalline A du composé de formule (I) préparée par le procédé tel que défini dans la revendication 10 pour une utilisation comme inhibiteur de β-lactamase pour traiter une infection par des bactéries productrices de carbapénèmase.

13. Forme cristalline A du composé de formule (I) telle que définie dans l'une quelconque des revendications 1-9 ou forme cristalline A du composé de formule (I) préparée par le procédé tel que défini dans la revendication 10 pour une utilisation comme inhibiteur de β-lactamase pour traiter une infection par *Klebsiella Pneumoniae* produisant la carbapénèmase KPC-2, *Klebsiella Pneumoniae* produisant la carbapénèmase NDM-1, ou *Klebsiella Pneumoniae* produisant la carbapénèmase OXA-181.

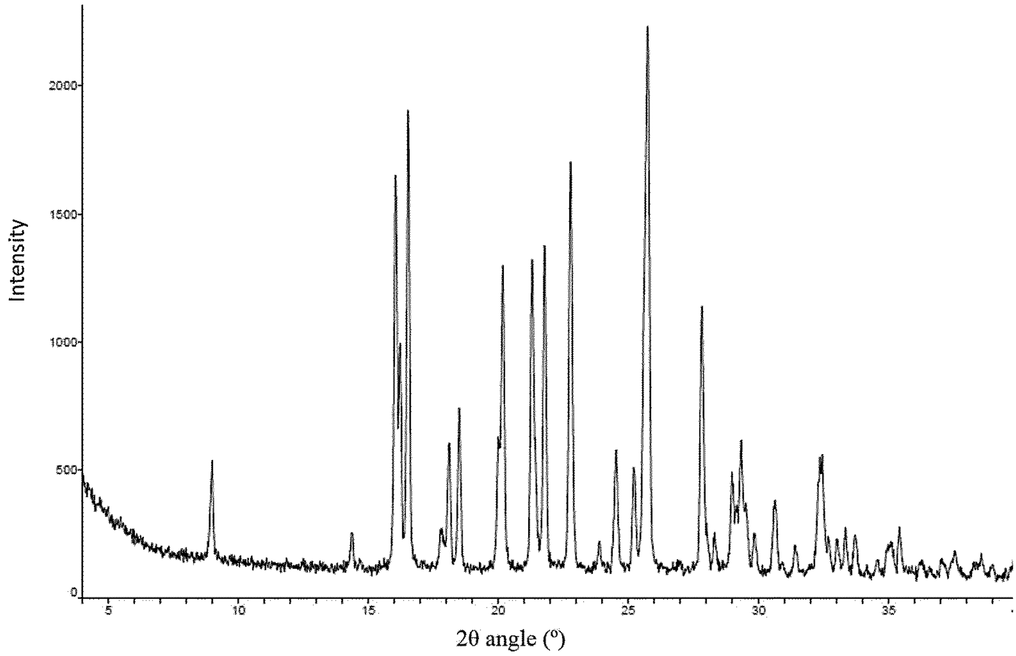

Intensity
2000
1500
1000
500
0
5
10
15
20
25
30
35
2θ angle (º)

Fig. 1

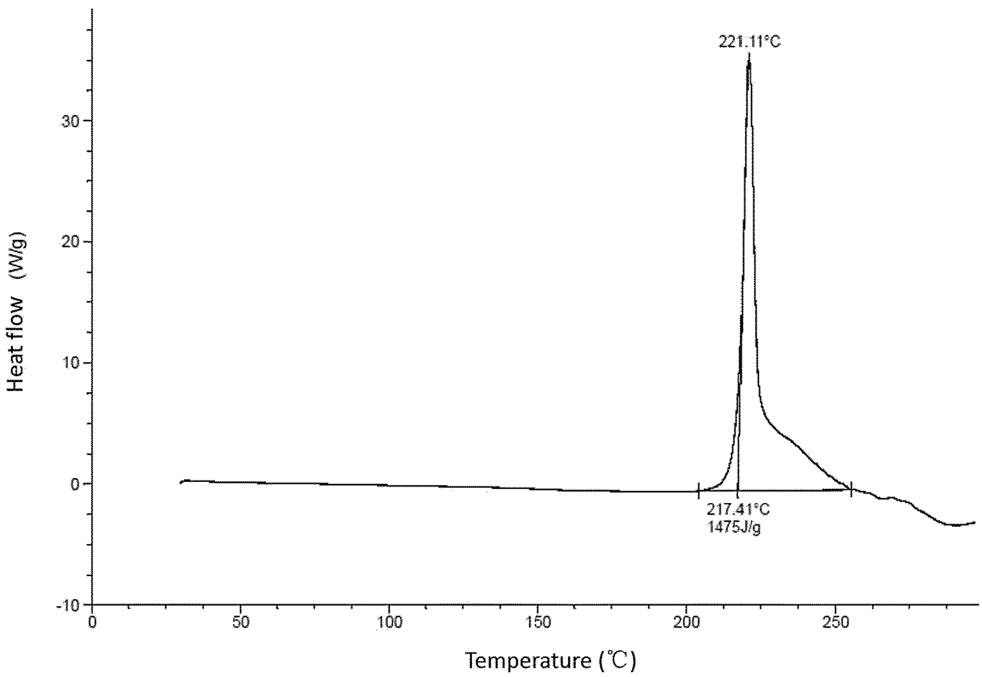

221.11°C
30
20
10
0
-10
Heat flow (W/g)
217.41°C
1475J/g
0
50
100
150
200
250
Temperature (℃)

Fig. 2